Autonomous AI Agents & Multi-Agent Workflows

ReAct Framework, Memory Stores, Tool Use & Orchestration

SkillForge AI Engineering Series - Certified Curriculum Materials

1. The ReAct (Reason + Act) Loop Pattern

Thought: The agent verbalizes reasoning about the current sub-goal.

Action: The agent calls an external tool Search, Database, Calculator, Python REPL.

Observation: The agent reads tool execution output and updates internal state.

2. Agent Memory & State Persistence

Short-Term Memory: In-context conversational buffer and scratchpad traces.

Long-Term Memory: Vector-indexed past episodes and user preferences.

Semantic Cache: Fast lookup for frequent tool queries to reduce LLM latency and cost.

3. Multi-Agent Collaboration Patterns

Hierarchical Orchestrator: Supervisor agent delegates tasks to specialized worker subagents.

Peer Critique: Evaluator agent critiques generated outputs against quality criteria.